

Outline

Pubertal Development

- Review literature on pubertal development in boys and girls and links to outcomes in adolescence, early adulthood, and mid-life.
 - Pubertal timing (i.e., the onset of puberty) demonstrates significant individual variability, with far reaching consequences for developmental processes.
 - In particular, early maturing girls have higher risk for mood disorders and externalizing behaviors such as delinquency, substance use, early sexual activity.
 - Both early- and late-maturing boys demonstrate elevated risk for depression and other internalizing behaviors
 - Comparatively, relatively little is known about the long-term impact of pubertal processes beyond adolescence and early adulthood but:
 - Some evidence suggests that links between pubertal timing and adverse outcomes are minimal by the fourth decade of life (Boden, 2011; Senia, 2018)
 - Other evidence suggests that risk conferred by early or off-time pubertal development persists into adulthood in certain domains (e.g., depression, sleep; Hoyt, 2020)
 - Understanding these links between pubertal processes and behavioral outcomes depends on measurement

The Measurement of Pubertal Timing

- What measures are available to measure pubertal timing?
 - Menarche (single indicator), self-report (e.g., PDS, peer relative timing measures), hormone measurements, physical exams
 - Can vary in terms of their objectivity (vs. subjectivity), whether they're concurrent or retrospective, or whether they're cross-sectional or longitudinal
 - Different measures have different strengths
 - Adolescents aren't great at identifying early pubertal signs which can lead to regression through the stages (Tanner Stage 3 → Tanner Stage 2)
 - Some items in the PDS aren't as reliable (e.g., skin changes; [Shirtcliff, 2009](#))
 - Timing might be best assessed after puberty. Explain why.
- Retrospective pubertal measures
 - There are several measures available to assess retrospectively reported pubertal timing including retrospective PDS measures and a timing measure (Beltz & Berenbaum, 2013).
 - Psychometrically validated/assessed → Cronbach's alpha, ICC, correlations with recalled age for menarche & spermatarche, but not with prospective measurements or a separate dataset
 - Retrospective timing measures are a powerful tool.

- Work with these retrospective measurements has revealed important insights into the persistence of mood disorders, substance use, and behavioral problems
- But concerns about the use of these measures persist:
 - Reliability: high internal consistency at the time of assessment (see Beltz & Berenbaum, 2013), but what about test-retest reliability or reporting over time?
 - Validity: what is the role of recall bias → can't look at validity because there's no objective reports of pubertal development during adolescence

What's Known about the Reliability and Validity of Retrospective Reporting?

- Review what sexual initiation and substance use studies tell us about reliability of retrospective reporting as a benchmark. Some key issues include:
 - Length of recall
 - Sex differences
 - Off-time/Regression to the mean
- Review menarche/timing studies (i.e., what gaps does our study fill?)
 - Marie & Jacobsen (2017): High reproducibility of menarche with long length between initial and retrospective report.
 - But both time points were retrospective
 - Gilger et al (1991): substantial test-retest reliability, MZ intraclass correlations exceeded DZ intraclass correlations, magnitude of relations followed theoretical expectations of traits with genetic influences
 - If many retrospective reports are taken at 18, then those might good reports
 - But only measured in adulthood
 - And small sample size
 - General limitations of these studies:
 - Both time points were retrospective
 - Both time points were in adolescence
 - Only menarche → no boys were assessed

The Current Study

- Compare prospective reports of status to a validated measure of retrospective pubertal timing using a large cohort of youth followed prospectively into adulthood
 - Aim 1: Examine the correspondence between prospective and retrospective reports of timing in boys and girls.
 - Compare these findings to sexual initiation
 - Aim 2: Examine whether prospective and retrospective reports of timing are differentially associated with outcomes.
 - Aim 3: Examine factors (moderate the accuracy of outcomes) associated with better or worse recall of outcomes (e.g., age, length of recall...)

Methods

Participants
Measures
Analytic Plan

Results

Aim 1: Examine the correspondence between prospective and retrospective reports of timing in boys and girls.

- Report means and zero order correlations for indicator (sexual initiation, menarche, girls’ timing, boys’ timing) in text
- Figure 1a & 1b: Average sexual initiation and menarche by early, on-time, and late categories
- Figure 2c & 2d: Average sexual girls’ and boys’ timing by early, on-time, and late categories
- Figure 3. Correspondence between prospective and retrospective reports of timing.

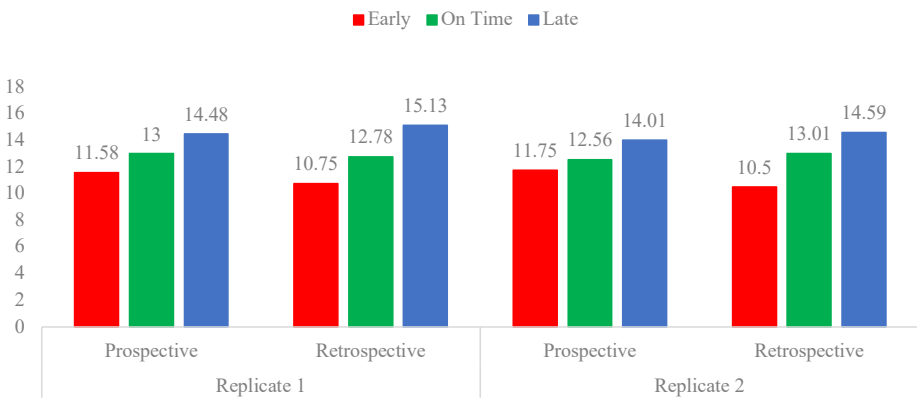


Figure 1b. EXAMPLE menarche by early, on-time, and late categories

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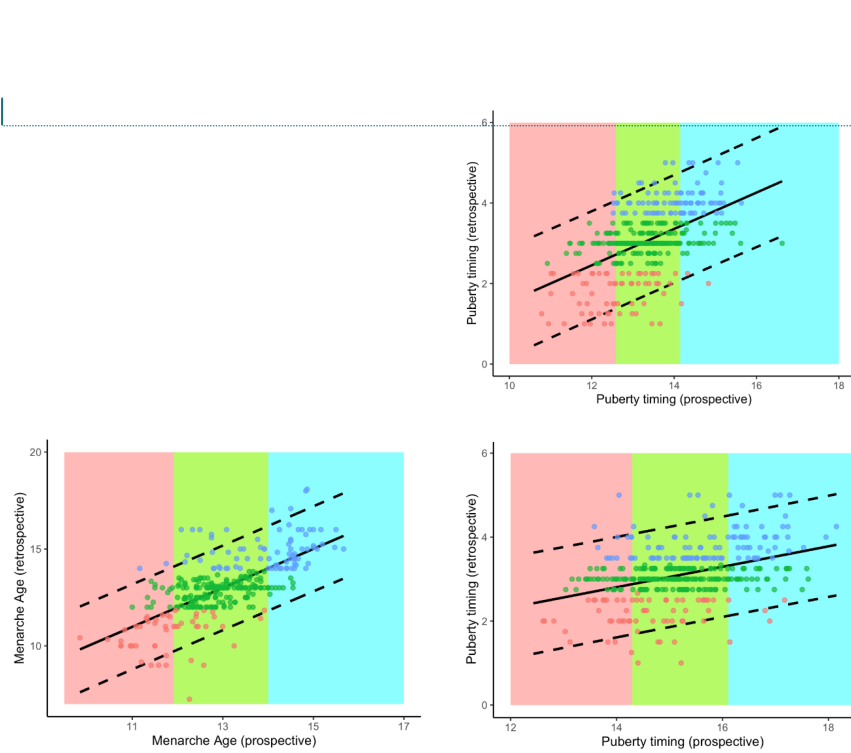


Figure 3. EXAMPLE Correspondence between prospective and retrospective reports of timing.

Commented [NC2]: girls' timing

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Aim 2: Examine whether prospective and retrospective reports of timing are differentially associated with outcomes.

- Figure 4. Depict outcomes using area plots (e.g., Beltz, 2014)

Aim 3: Examine factors (moderate the accuracy of outcomes) associated with better or worse recall of outcomes (e.g., age, length of recall...)

- Report in text

Discussion

